

A Clifford Construction Answering Problem 5.9 in *Phase Retrieval by Projections*

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Source problem

Source: *Phase Retrieval by Projections*, arXiv:1305.6226 (first posted May 27, 2013).

Authors: Jameson Cahill, Peter G. Casazza, Jesse Peterson, and Lindsey Woodland.

The authors ask for examples of subspaces that allow phase retrieval although the span of their associated projections is not the span of any equally large family of rank-one projections [1]. This is Problem 5.9 on page 16 of the arXiv PDF (the source labels it P5).

$z^2 = \frac{1}{5}$. Then the vectors in $\mathbb{R}^3$ given by $\{(x_{i1}, x_{i2}, z)\}_{i=1}^5$ form a full spark Parseval frame for $\mathbb{R}^3$ and hence the above example works.

Remark 5.8. The above example can be generalized to Parseval fusion frames. However, much less is known about Parseval fusion frames with arbitrary dimensional subspaces. We do not want to introduce this subject here and this will be covered in a later paper.

The following problem is also open at this time.

Problem 5.9. Find examples of (classify) the subspaces $\{W_n\}_{n=1}^N$ which allow phase retrieval but the span of their associated projections $\{P_n\}_{n=1}^N$ is not equal to the span of any N rank one projections.

Result

This packet gives a **candidate full solution**: seven explicit two-dimensional subspaces of $\mathbb{R}^4$ allow phase retrieval, while the seven-dimensional span of their orthogonal projections contains no nonzero rank-one matrix. The latter conclusion is stronger than the requirement in the problem.

Here “allows phase retrieval” means that the map

$$x \in \mathbb{R}^4 / \{\pm 1\} \mapsto (\|P_1 x\|^2, \dots, \|P_7 x\|^2)$$

is injective. Equivalently, equality of all seven measurements forces $x = \pm y$.

Theorem 1. There are seven rank-two orthogonal projections $P_1, \dots, P_7$ on $\mathbb{R}^4$ such that

1. $\{\text{ran } P_i\}_{i=1}^7$ allows phase retrieval;
2. $\text{span}\{P_1, \dots, P_7\}$ contains no nonzero matrix of rank one.

Consequently this span is not the span of any seven rank-one projections.

Idea of the proof

Choose a three-dimensional subspace $K \subset \text{Sym}_4(\mathbb{R})$ generated by three pairwise anticommuting symmetric involutions. The Clifford relations make every nonzero member of K invertible. Its orthogonal complement $S = K^\perp$ is then the span of seven explicit rank-two projections. The

phase-retrieval obstruction $xx^\top - yy^\top$ has rank at most two, so it cannot lie nontrivially in the measurement kernel K .

The remaining requirement is more delicate: S must not be generated by rank-one projections. In fact it contains none. Orthogonality of $zz^\top$ to the three Clifford generators produces the three quadratic coordinates of the real Hopf map; their sum-of-squares identity forces $z = 0$.

Construction

Put

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

All tensor products below act on $\mathbb{R}^2 \otimes \mathbb{R}^2 = \mathbb{R}^4$, in the standard lexicographic basis. Define

$$A_1 = Z \otimes I, \quad A_2 = X \otimes I, \quad A_3 = Y \otimes Y$$

and

$$\begin{aligned} Q_1 &= I \otimes X, & Q_2 &= I \otimes Z, & Q_3 &= X \otimes X, \\ Q_4 &= X \otimes Z, & Q_5 &= Z \otimes X, & Q_6 &= Z \otimes Z. \end{aligned}$$

Let I_4 denote the identity on $\mathbb{R}^4$, and set

$$P_1 = \frac{I_4 + Q_1}{2}, \quad P_2 = \frac{I_4 - Q_1}{2}, \quad P_{j+1} = \frac{I_4 + Q_j}{2} \quad (2 \leq j \leq 6).$$

Each Q_j is a symmetric traceless involution. Hence every P_i is an orthogonal projection of rank two.

For completeness, the seven requested subspaces can be read without tensor notation from the following table. Scaling the displayed generators is irrelevant.

i	$\text{ran } P_i$
1	$\text{span}\{(1, 1, 0, 0), (0, 0, 1, 1)\}$
2	$\text{span}\{(1, -1, 0, 0), (0, 0, 1, -1)\}$
3	$\text{span}\{(1, 0, 0, 0), (0, 0, 1, 0)\}$
4	$\text{span}\{(1, 0, 0, 1), (0, 1, 1, 0)\}$
5	$\text{span}\{(1, 0, 1, 0), (0, 1, 0, -1)\}$
6	$\text{span}\{(1, 1, 0, 0), (0, 0, 1, -1)\}$
7	$\text{span}\{(1, 0, 0, 0), (0, 0, 0, 1)\}$

Proof

Equip $\text{Sym}_4(\mathbb{R})$ with the Hilbert–Schmidt inner product $\langle B, C \rangle = \text{tr}(BC)$. The ten matrices

$$I \otimes I, I \otimes X, I \otimes Z, X \otimes I, X \otimes X, X \otimes Z, Z \otimes I, Z \otimes X, Z \otimes Z, Y \otimes Y \quad (1)$$

are nonzero, symmetric, and pairwise Hilbert–Schmidt orthogonal. Since $\dim \text{Sym}_4(\mathbb{R}) = 10$, they form an orthogonal basis.

Let

$$K = \text{span}\{A_1, A_2, A_3\}, \quad S = K^\perp.$$

Reading off the basis (1) gives

$$S = \text{span}\{I_4, Q_1, \dots, Q_6\}. \quad (2)$$

Moreover,

$$I_4 = P_1 + P_2, \quad Q_1 = P_1 - P_2, \quad Q_j = 2P_{j+1} - I_4 \quad (2 \leq j \leq 6).$$

Thus

$$\text{span}\{P_1, \dots, P_7\} = S. \quad (3)$$

The matrices A_1, A_2, A_3 are symmetric involutions which pairwise anticommute. It follows that, for all $a, b, c \in \mathbb{R}$,

$$(aA_1 + bA_2 + cA_3)^2 = (a^2 + b^2 + c^2)I_4. \quad (4)$$

In particular, every nonzero member of K is invertible and has rank four.

We now prove phase retrieval. Suppose $x, y \in \mathbb{R}^4$ satisfy $\|P_i x\|^2 = \|P_i y\|^2$ for every i . With $D = xx^\top - yy^\top$, this says

$$\langle P_i, D \rangle = 0 \quad (1 \leq i \leq 7).$$

By (3), $D \in S^\perp = K$. But $\text{rank } D \leq 2$, whereas (4) says that every nonzero element of K has rank four. Therefore $D = 0$. Hence $xx^\top = yy^\top$, which implies $x = \pm y$. The seven subspaces allow phase retrieval.

It remains to prove the stronger rank-one exclusion. If $zz^\top$ lies in $S = K^\perp$, then $z^\top A_j z = 0$ for $j = 1, 2, 3$. Writing $z = (z_1, z_2, z_3, z_4)$, these three quadratic forms are

$$\begin{aligned} q_1(z) &= z_1^2 + z_2^2 - z_3^2 - z_4^2, \\ q_2(z) &= 2(z_1 z_3 + z_2 z_4), \\ q_3(z) &= 2(z_1 z_4 - z_2 z_3). \end{aligned}$$

They satisfy the elementary Hopf identity

$$q_1(z)^2 + q_2(z)^2 + q_3(z)^2 = (z_1^2 + z_2^2 + z_3^2 + z_4^2)^2. \quad (5)$$

Thus $q_1(z) = q_2(z) = q_3(z) = 0$ forces $z = 0$. So S contains no nonzero rank-one matrix.

Finally, if S were the span of seven rank-one projections $u_i u_i^\top$, every one of those nonzero generators would itself belong to S , contradicting the preceding paragraph. This proves both claims of Theorem 1. $\square$

Machine-checkable audit

The accompanying exact SymPy script checks symmetry, involution and anticommutation; the determinant identity

$$\det(aA_1 + bA_2 + cA_3) = (a^2 + b^2 + c^2)^2;$$

orthogonality and dimensions of K, S ; idempotence and rank of all seven projections; equality of their span with S ; rank of the measurement map; and the polynomial identity (5). No floating-point computation is used.

Novelty note

A bounded search of the local corpus and arXiv/web indexes found later work answering other projection-phase-retrieval questions, including generic prescribed-rank constructions [2] and further structural results [3], but no source presenting this example or explicitly resolving Problem 5.9. This is a research packet, not a priority claim; specialist bibliographic review remains necessary.

Limitations and review recommendation

The result is an existence example in $\mathbb{R}^4$, as requested; it does not classify all such families or address the final complex minimality question in the source paper. The proof is finite-dimensional and exact, and every matrix identity has been independently checked by symbolic computation. Promotion for specialist review is recommended, with particular attention to the span identification (3) and the rank-one exclusion (5).

References

- [1] J. Cahill, P. G. Casazza, J. Peterson, and L. Woodland, *Phase Retrieval by Projections*, arXiv:1305.6226.
- [2] D. Edidin, *Projections and Phase Retrieval*, arXiv:1506.00674.
- [3] J. Cahill, P. G. Casazza, J. Jasper, and L. M. Woodland, *Phase Retrieval*, arXiv:1508.06687.